

SMPC Mini-Project 2

Use the buck converter you have built from mini-project 1, and build a control loop around it. You can use an ideal reference voltage of 1.0V, ideal amplifiers if needed. The control loop supply voltage is 5.0V. You have a PWM carrier with a peak-to-peak amplitude of 4.0V.

1. What is the LC cut-off frequency?
2. Assume the LC components have 20% tolerance, what is the range of possible LC cut-off frequency?
3. Design a voltage control loop to ensure stability (PM >60 degree), and to achieve a loop gain of at least 40 dB @ 200 Hz, the higher the better. Motivate your design choice. State the pole and zero frequencies. Try to avoid using extremely large passive components such as resistors > 10 Mohm, or capacitors > 100 pF.
4. Build an AC test bench to run stability analysis. Give the bode plot and annotate the loop gain @ 200 Hz, PM and control loop unity gain frequency.
5. In transient simulation, check the converter and compensator output waveform to see whether the system is indeed stable.
6. Add a peak-to-peak 1V sinewave on top of the input voltage, check the output of the converter waveform. Try to extract the component at 200 Hz. What do you observe and explain?